Sanjita Chandan Ballapur

(347) 835-9183 • sb5216@columbia.edu • linkedin.com/in/sanjitacballapur • github.com/sanjita811

Education

- **Columbia University, Columbia Engineering** — New York, NY
  M.S. in Computer Science, Machine Learning Track; GPA: 3.48/4.0 — *Expected Dec 2026*
  ***Courses**: High Performance Machine Learning, Agentic Systems, Algorithms, Distributed Systems, Natural Language Processing*

- **PES University** — Bangalore, IN
  Bachelor of Technology in Computer Science and Engineering; GPA: 9.69/10.0 — *May 2024*
  ***Courses:** Python, Data Structures & Algorithms, Artificial Intelligence, Cloud, Operating Systems, C, C++ Java, Databases*

Experience

- **IBM Research** — Yorktown Heights, NY
  Position: AI Systems Engineer Intern — *May 2026 – Aug 2026*
  - Implemented the Mamba-2 State Space Duality (SSD) kernel for IBM's Spyre AI accelerator in the open-source torch-spyre PyTorch backend, laying the foundation for efficient long-context models (Codestral, Granite 4, and Nemotron) on IBM's Spyre AI accelerators; validated against a CPU version with 0.4% relative L2 error under FP16 precision
  - Redesigned sequential math as matrix multiplications and split the work across parallel cores, removing CPU fallbacks and staying within per-core memory limits
  - Profiled the kernel through the compiler stack and found it was memory-bound; removed duplicated tensor re-layouts, cutting 11% of device time
  - Built a three-tier validation harness so failures could be isolated to the algorithm, the numerics, or the compiler. Leveraged Claude Code to keep the kernel in sync with fast-moving upstream changes and iterate on parallelization and memory-layout optimizations, verifying each change on hardware

- **Hewlett Packard Enterprise (HPE)** — Bangalore, IN
  Position: College Intern & Data Engineer — *Feb 2024 – Aug 2024 & Sep 2024 – Jun 2025*
  - Integrated protobuf support into Spark/Kafka pipelines, enabling faster serialization/deserialization of structured messages and reducing compute costs by 20% by eliminating Avro conversion overhead
  - Led the end-to-end migration of software applications from cloud to on-premises clusters to meet security requirements, re-architecting data pipelines and refactoring the codebase for seamless multi-environment compatibility
  - Designed and implemented a structured storage strategy for compacted Kafka topics using Delta Lake instead of Parquet, reducing latency and cutting storage costs by 50%

- **Troogue.ai** — Bangalore, IN
  Position: Machine Learning Intern — *Jun 2023–Sep 2023*
  - Built an ML/NLP system for resume analysis, developing datasets and training models like BERT, Logistic Regression, and Random Forest to extract and rank technical and soft skills, improving candidate-job matching accuracy by 30%
  - Productionized ML models as scalable API endpoints using microservices, collaborating with cross-functional teams to integrate the system into the broader application

Key Academic Projects

- **Uncertainty-Aware Inference for Quantized LLMs (Collaboration with IBM Research)**: Benchmarked Mistral-7B under four quantization methods (GPTQ INT4/INT8, AWQ, NF4) across three datasets, measuring inference speed and reliability of confidence scores. Profiled GPU kernels with PyTorch Profiler, Nsight Compute, and Roofline analysis; showed kernel choice, not bit-width, drives speed: GPTQ INT4 (Marlin) reached 1.78× FP16 throughput under vLLM, while AWQ INT4 at the same precision ran 4.5× slower. Found that quantization can improve average calibration error (0.229 → 0.067) while worst-case error rises (0.33 → 0.94) on a medical QA benchmark. Built a confidence-based routing analysis showing that sending low-confidence queries to FP16 saves cost only when the quantized kernel is faster under batched serving.
- **Low-Resource Machine Translation from Ancient Egyptian to German**: Fine-tuned LLaMA-3.1-8B-Instruct using LoRA for low-resource Ancient Egyptian-to-German translation, achieving 45.62 BLEU and 61.62 chrF, with minimum Bayes risk decoding. Extended the tokenizer with domain-specific Egyptian transliteration tokens learned via byte pair encoding to improve out-of-distribution generalization. Ran ablations across model scale, tokenization, input representation, and decoding strategy, with a multilingual M2M-100 baseline.
- **Real-Time De-escalation Engine for Conversational LLMs**: Built a closed-loop system that detects rising user frustration and adapts an LLM chatbot's tone to prevent escalation, using GoEmotions BERT + VAD projection for emotion tracking and a rolling-window XGBoost model to predict escalation 2 turns ahead (AUC 0.79 on 50K conversations). Applied SHAP to convert top escalation drivers into natural-language directives injected into a quantized Mistral-7B system prompt, validated via a paired A/B test showing a statistically significant +0.27 mean valence shift ($p < 0.05$) toward warmer replies.

Skills

- **Languages**: Python, Java, Scala, C, C++, SQL, HTML, Javascript
- **ML/AI**: PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, XGBoost, vLLM, Keras, Numpy, Pandas, NLTK
- **Data & Distributed Systems**: Spark, Kafka, Hadoop, Airflow, Delta Lake
- **Systems & Cloud**: Docker, Kubernetes, Git, AWS, Google Cloud Platform
- **AI-Assisted Development**: Claude Code, GitHub Copilot, Codex